

Moist V1: gross dry stability, mean DSE flux, and gross moist stability, with the equilibrium MSE-flux decomposition

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This spells out, from the spec (`guides/moist_axisymmetric_model_spec`), the definitions of $\hat{S}$, the mean DSE flux, and $\hat{H}$, evaluates them at the V1 equilibrium (`ny=801`, $\Delta t=30$, $\sin^2 y_0=0$, slow-drift-gated, day 1855, $D=10^6$), and resolves why the mean flow carries a small *equatorward* MSE flux while its DSE part is poleward.

1 Column accounting

Write the layer thermodynamic and moisture equations of V1 (spec Eqs. T1, Q1):

$$\partial_t \theta + \frac{d\Delta_z}{H} \partial_y v = \frac{\theta_E - \theta}{\tau}, \quad \partial_t W + \partial_y [-(2a-1)vW - D \partial_y W] = E_0 - P. \quad (1)$$

The column heat capacity that turns a θ -tendency into a column-energy tendency, and the column dry static energy, are

$$C \equiv \frac{c_p \Pi \Delta p}{g}, \quad \langle s \rangle \equiv C\theta. \quad (2)$$

With the SS13 Exner factor $\Pi = 1/1.6 = 0.625$, $\Delta p = p_s - p_t = 807$ hPa, and $c_p = 1004$ J kg⁻¹ K⁻¹,

$$C = \frac{1004 \times 0.625 \times 8.07 \times 10^4}{9.81} \approx 5.2 \times 10^6 \text{ J m}^{-2} \text{ K}^{-1}. \quad (3)$$

The latent-heating constant is pinned to the *same* C by requiring $C\Lambda_{\text{conv}} = L_v$, i.e. $\Lambda_{\text{conv}} = L_v/C \approx 0.48$ K/(kg m⁻² s⁻¹) for $L_v = 2.5 \times 10^6$ J kg⁻¹.

2 Gross dry stability and the mean DSE flux

Multiply the thermodynamic equation by C . Since C and the coefficient $d\Delta_z/H$ are constant, the adiabatic term becomes a flux divergence:

$$\partial_t \langle s \rangle + \partial_y \left[\underbrace{C \frac{d\Delta_z}{H}}_{\hat{S}} v \right] = \frac{C}{\tau} (\theta_E - \theta). \quad (4)$$

So the **gross dry stability** is $\hat{S} \equiv C d\Delta_z/H$ and the **mean DSE flux** is $\hat{S}v$. Numerically, with the model's $d = 4000$ m, $\Delta_z = 60$ K, $H = 1.6 \times 10^4$ m,

$$\frac{d\Delta_z}{H} = \frac{4000 \times 60}{16000} = 15 \text{ K}, \quad \boxed{\hat{S} = C \frac{d\Delta_z}{H} = 5.2 \times 10^6 \times 15 = 7.8 \times 10^7 \text{ J m}^{-2}} \quad (5)$$

$\hat{S} > 0$: a poleward v (upper branch, $v > 0$ in the NH) exports dry static energy poleward, cooling the deep tropics, exactly as expected.

3 Gross moist stability and the MSE budget

Add $L_v \times (\text{moisture equation})$ to $C \times (\text{thermodynamic equation})$. Using $C\Lambda_{\text{conv}} = L_v$ the latent heating $\Lambda_{\text{conv}}P$ and the precipitation sink L_vP cancel, leaving a pure column-MSE budget with $\langle h \rangle = C\theta + L_vW$:

$$\partial_t \langle h \rangle + \partial_y \left[\underbrace{\hat{S}v}_{\text{DSE}} - \underbrace{L_v(2a-1)Wv}_{L_vq} - \underbrace{L_vD\partial_yW}_{\text{eddy}} \right] = \frac{C}{\tau}(\theta_E - \theta) + L_vE_0. \quad (6)$$

The mean (advective) part groups into $v\hat{H}$ with the **gross moist stability**

$$\boxed{\hat{H} \equiv \hat{S} - L_v(2a-1)W} \quad (7)$$

At the equilibrium equatorial $W \approx 50.76 \text{ kg m}^{-2}$ and $a = 0.85$ (so $2a - 1 = 0.7$),

$$\hat{H} = 7.8 \times 10^7 - (2.5 \times 10^6)(0.7)(50.76) = 7.8 \times 10^7 - 8.88 \times 10^7 = -1.08 \times 10^7 \text{ J m}^{-2}. \quad (8)$$

$\hat{H} < 0$: the latent term outweighs the dry stability, the moisture-mode side. The zero crossing is at $W = \hat{S}/[L_v(2a-1)] = 44.6 \text{ kg m}^{-2}$, which the column exceeds everywhere, so $\hat{H} < 0$ across the whole domain.

4 Why the net mean MSE flux is small and equatorward

The mean MSE flux $\hat{S}v - L_v(2a-1)Wv = v\hat{H}$ is the *difference* of two large opposing fluxes. At the cell center ($|y| \approx 2.9$ Mm, $|v| = 0.365 \text{ m s}^{-1}$):

term	MW m ⁻¹	direction
DSE $\hat{S}v$	28.5	poleward
$L_vq - L_v(2a-1)Wv$	32.4	equatorward
mean MSE $v\hat{H}$	3.9	equatorward (residual)
eddy $-L_vD\partial_yW$	$\lesssim 0.15$	negligible

The dry circulation exports 28.5 MW m^{-1} of DSE poleward; the moisture flux, carried mainly by the equatorward lower branch, returns 32.4 MW m^{-1} equatorward. Their small residual, 3.9 MW m^{-1} up-gradient toward the ascending ITCZ, is the net MSE flux and the signature of $\hat{H} < 0$. Plotting only $v\hat{H}$ (as the first diagnostic did) hides that it is the difference of two $\sim 30 \text{ MW m}^{-1}$ fluxes; the implied L_vq flux is indeed large.

5 The overturning is eddy-driven

The overturning that sets these fluxes is weak, $|v|_{\max} = 0.37 \text{ m s}^{-1}$, and follows the eddy-driven leading balance of the zonal-momentum equation, Coriolis against the eddy momentum flux divergence:

$$\beta y v \approx v_d \mathcal{H}(u) \operatorname{sgn}(y) \partial_y u \equiv \text{EMFD}, \quad v \approx \frac{\text{EMFD}}{\beta y}. \quad (9)$$

At the cell center $\beta y v = 2.1 \times 10^{-5}$ balances $\text{EMFD} = 1.8 \times 10^{-5} \text{ m s}^{-2}$ with Rayleigh drag $\varepsilon_u u = 8.5 \times 10^{-8}$ two orders of magnitude smaller; the $\sim 13\%$ gap is the nonlinear advection. The overturning strength therefore scales with v_d (fixed at 2.5 across this D ladder).